

Project Proposal

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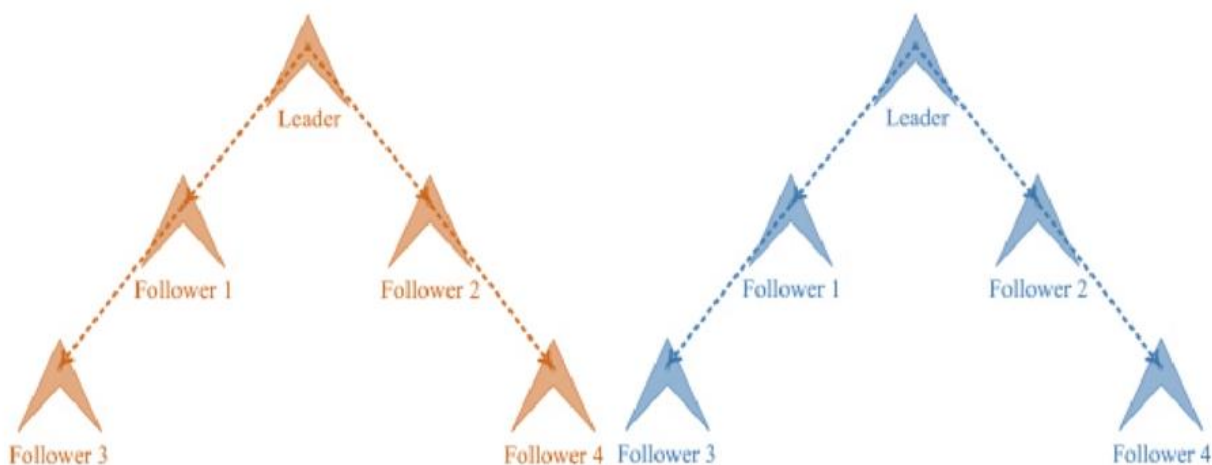
Topic: A Secure and Fault-Tolerant Drone Swarm System with Automatic Leader Replacement

1. What I Am Trying to Do

In this project, I want to build a team of drones that can work together intelligently instead of depending on a single fixed controller or leader drone.

Normally, in multi-drone systems, one drone acts as the leader. If that leader drone crashes or loses communication, the whole system stops working. I want to solve this problem by designing a system where:

- One drone acts as a leader
- Other drones follow and cooperate with it
- All drones communicate with each other internally
- If the leader drone fails, the remaining drones automatically choose a new leader
- The mission continues without human intervention



The drones will also be able to make basic autonomous decisions using machine learning, such as obstacle avoidance and maintaining formation.

2. Why I Want to Do This (Problem Motivation)

Most existing drone swarm systems have these limitations:

- Dependence on a fixed leader drone
- Leader failure causes mission failure
- Communication is often centralized or insecure
- Little or no autonomy in decision-making

In real-life applications like search-and-rescue, disaster monitoring, or surveillance, stopping a mission because one drone fails is unacceptable. This motivates me to design a fault-tolerant and self-recovering drone swarm system.

3. What Exactly I Want to Build (Project Objective)

The main goals of this project are:

- Build a multi-drone system where drones work as a team
- Allow only one drone to act as a leader at a time
- Enable automatic leader replacement if the leader fails
- Establish secure internal communication among drones
- Use machine learning for autonomous decision support
- Test the system using simulation and/or real drones

4. Overall System Concept (How the System Works)

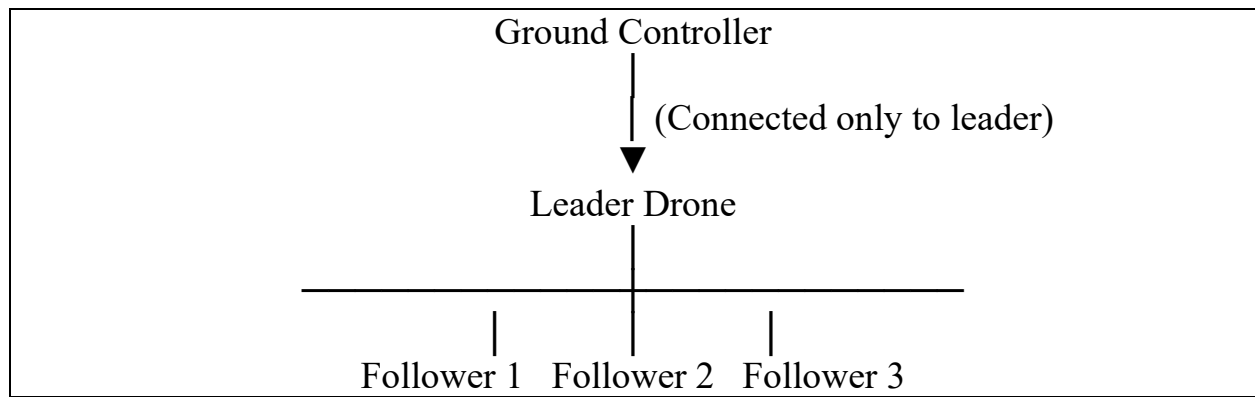
Normal Operation

- One drone becomes the leader
- The leader communicates with the ground controller
- Other drones follow the leader and share status information

Failure Situation

- If the leader drone crashes or stops responding
- Other drones detect the failure using heartbeat signals
- A leader election process starts
- The most suitable drone becomes the new leader
- The mission continues automatically

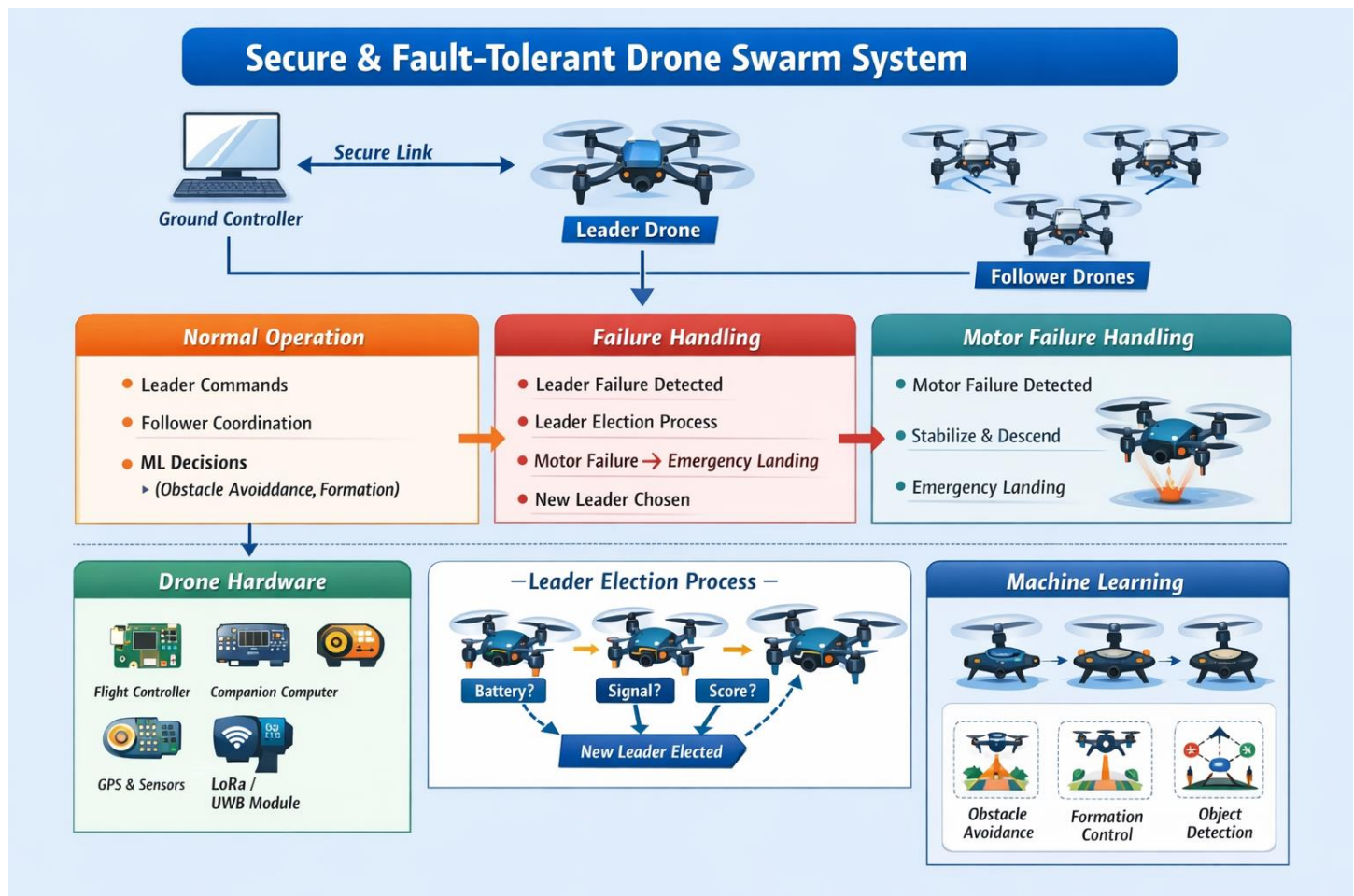
5. System Architecture Diagram



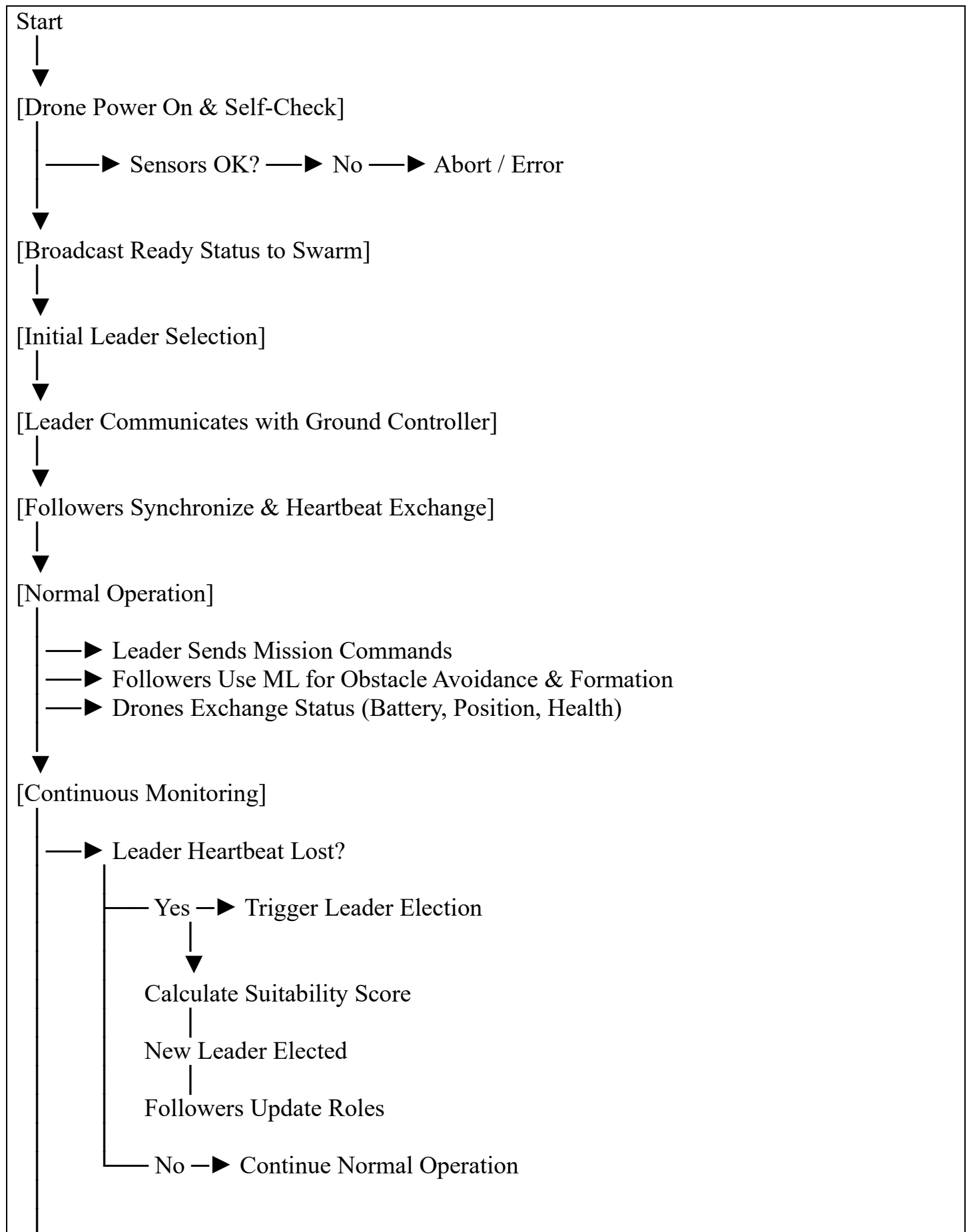
Internal Drone Structure

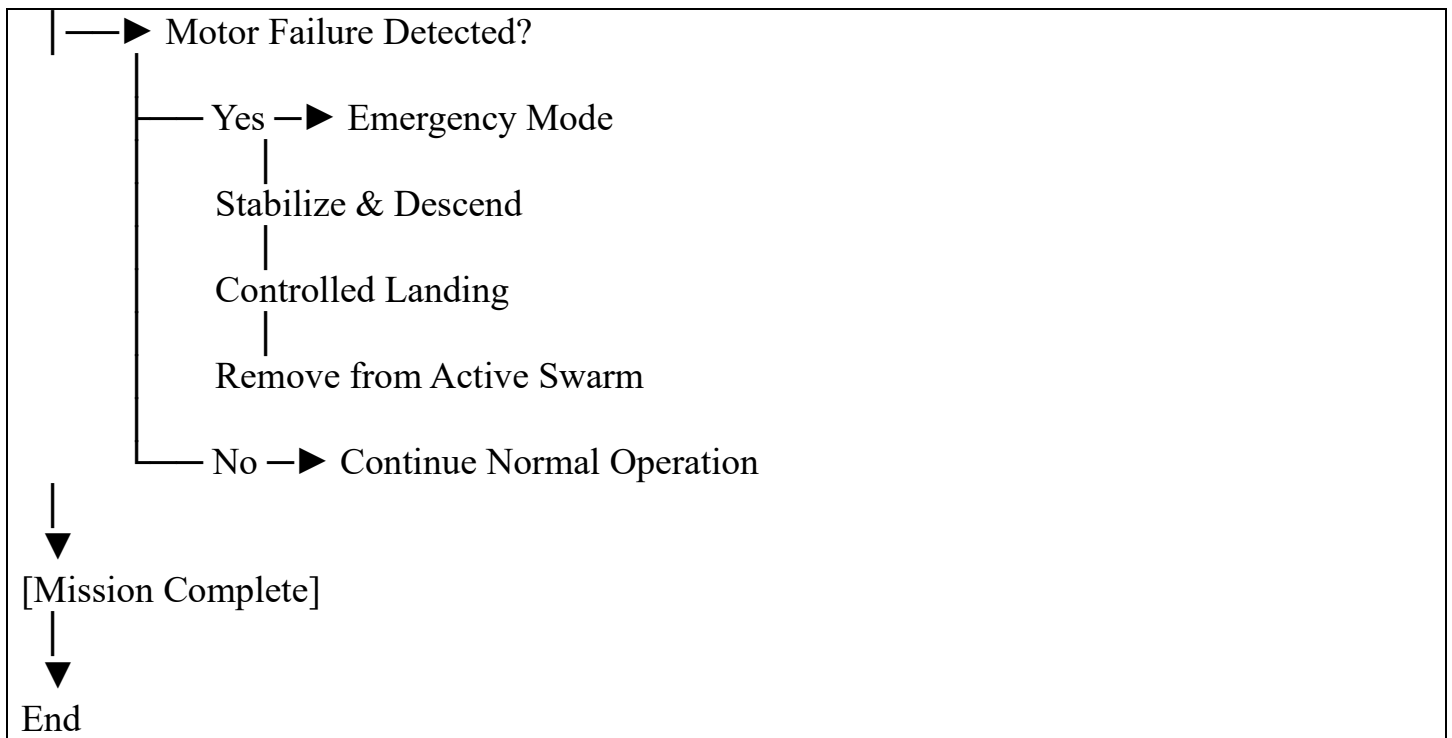


Drone swarm system infographic diagram



Drone Swarm System Workflow Diagram





6. Hardware Components

Each drone will contain:

- Flight Controller (PX4 / ArduPilot)
- Companion Computer (Raspberry Pi or Jetson Nano)
- GPS and IMU
- Distance sensors (LiDAR or ultrasonic)
- Wireless module (LoRa / NRF / UWB)

No new drone design is required; existing drones will be extended.

7. Software Components

- PX4 Autopilot firmware
- MAVLink / MAVSDK for communication
- Python-based swarm control logic
- Secure communication module (AES encryption)
- Machine learning inference module

8. Secure Communication Approach

- Drones communicate directly with each other
- No IP address or internet is required
- Messages are encrypted
- Each drone sends periodic heartbeat signals
- Only authenticated drones can join the swarm

9. Leader Election Method

When the leader drone fails:

1. Other drones detect missing heartbeat
2. Each drone calculates its own suitability score based on:
 - Battery level
 - Signal strength
 - Processing capability
3. Drones exchange scores
4. The drone with the highest score becomes the new leader
5. Other drones update their role automatically

10. Motor Failure Handling (Fault-Tolerance)

In addition to leader failure, this system considers partial hardware failure, specifically motor failure in quadcopter drones.

- Standard quadcopters cannot continue normal flight if one motor fails
- Instead of attempting continued flight, the drone enters emergency mode
- Remaining motors stabilize orientation as much as possible
- Drone performs a controlled emergency landing
- The damaged drone exits swarm operation
- Remaining drones automatically reorganize and continue the mission

Future Improvement: Using hexacopters or octocopters could allow partial flight even after a motor failure.

11. Use of Machine Learning

Machine learning is used for decision support, not for critical flight control.

ML will assist in:

- Obstacle avoidance
- Formation maintenance
- Movement optimization

Safety-critical flight control is handled by the flight controller firmware.

12. Programming Languages and Tools

Languages

- Python: For swarm logic, leader election, communication, and ML inference
- C/C++: For PX4 flight control, low-level sensor and motor management

Tools / Frameworks

- PX4 Autopilot
- MAVSDK / MAVLink
- ROS2 (optional, for simulation)
- PyTorch / TensorFlow / TinyML for ML model deployment
- Gazebo / SITL for simulation
- Python cryptography library for secure communication

13. Development Methodology

1. Design system architecture
2. Implement simulation using PX4 SITL
3. Develop secure communication protocol
4. Implement leader election logic
5. Integrate machine learning module
6. Test leader and motor failure scenarios
7. Analyze performance and reliability

14. Expected Outcome

- A working drone swarm prototype
- Automatic leader recovery without human intervention
- Safe handling of motor failures with emergency landing
- Secure and stable drone-to-drone communication
- Demonstration of autonomous decision-making
- Scalable foundation for future research

15. Scope and Limitations

Scope

- Small swarm (2–5 drones)
- Simulation and limited real-world testing
- Academic research focus

Limitations

- Not designed for military-grade security
- Limited by onboard computing power
- Environmental constraints during testing

Project Intent and Expected Impact

This project aims to build a robust and intelligent drone swarm system that can survive both leader failure and partial hardware failures such as motor malfunction. Previous research has explored leader–follower swarm architectures, decentralized coordination, leader election mechanisms, secure drone communication, and the use of machine learning for navigation and obstacle avoidance. However, most existing works address these problems individually and are often limited to simulation or centralized control models. A fully integrated and practical system that simultaneously combines secure internal communication, automatic leader replacement, machine learning–based decision support, and emergency handling of partial hardware failure is still rare. The system proposed in this project does not directly replicate any existing solution but instead builds upon established research concepts and extends them into a unified experimental framework, making this work an original implementation and not a direct copy of prior research.